

Siyuan Jiang

 siyuanj |  jiang-sy23@mails.tsinghua.edu.cn |  188 1131 1636 |  siyuanj.github.io

EDUCATION

Tsinghua University, Tanwei College
B.S. in Chemical Biology for Pharmaceutical Science

2023 - 2027 (expected)
Beijing, China

- GPA: 3.83/4.00
- Selected Courses: *Chemical Bioinformatics and Artificial Intelligence* (A+), *Drug Design* (A+), *Bioinformatics* (A, Excellent Project), *Medicinal Chemistry* (A), *AI in Healthcare* (A), *Chemical Biology* (A-)

RESEARCH EXPERIENCE

Structure-Guided Aptamer Design

Jul 2026 - Present

Advisor: Prof. Le Cong

School of Medicine, Stanford University

- Developed workflows for RNA/DNA aptamer design, integrating protein–nucleic acid complex prediction, backbone-conditioned inverse folding, and structure-based candidate filtering.
- Benchmarked AlphaFold 3, Boltz-2, Protenix, and OpenFold3, evaluating nucleic acid folding, protein-relative binding poses, and interface quality to assess the reliability of predicted training structures.
- Evaluated NALite inverse-folding models with predicted and generated structural data under matched training budgets, comparing RNA/DNA sequence recovery and data-source effects.

EasyDesign: Agent-Assisted Protein Design

Jun 2026 - Present

Advisor: Assoc. Researcher Yafei Yuan

School of Life Sciences, THU

- Contributed across multiple stages of EasyDesign’s agent-assisted protein and binder design workflow.
- Primarily responsible for optimizing structure-prediction models and inference workflows; configured an inference backend using AlphaFold 3 code with converted OpenFold3 weights and packaged reproducible GPU environments for deployment on laboratory servers.

High-Throughput Screening of Functional Food-Derived Peptides

Oct 2025 - Oct 2026

Advisor: Assoc. Researcher Yafei Yuan

School of Life Sciences, THU

- Developed a multi-stage screening pipeline for bioactive food-derived peptides targeting specific proteins, addressing the computational bottleneck of direct high-precision docking.
- Constructed and trained a lightweight pre-screening model to efficiently filter massive peptide libraries and enrich potential candidates for downstream analysis.
- Integrated AlphaFold 3 for high-precision structural validation of top candidates and finalized the study with wet-lab experimental assays to verify binding affinity.

Spatiotemporal Evolution of the Gamma-Secretase Complex

Jul 2025 - Present

Advisor: Prof. Yigong Shi

School of Life Sciences, THU

- Constructed eukaryotic phylogenetic trees for the gamma-secretase complex and traced the PS1 subunit from prokaryotic homologs to clarify its deep evolutionary origin.
- Batch-predicted approximately 1,500 structures with AlphaFold 3 and analyzed the stepwise assembly order of gamma-secretase subunits during evolution.
- Applied statistical methods and deep learning tools to identify mechanistically important residues in gamma-secretase and explore their potential association with Alzheimer’s disease.

PUBLICATION

Siyuan Jiang, Sijin Wu, Chi Wang, Zhaorui Jiang, Dongjian Jiang, Yiquan Wang, Tin-Yeh Huang, Xu Wang, and Yafei Yuan. **GPCR-Turbo: Fast and Accurate Peptide Virtual Screening for GPCRs**. *2026 IEEE International Conference on Bioinformatics and Biomedicine (IEEE BIBM 2026)*. Accepted.

Xu Wang, Chi Wang, Tin-Yeh Huang, Yiquan Wang, **Siyuan Jiang**, and Yafei Yuan. **StrucTrace: A universal Fourier watermark for traceable biomolecular structures**. *npj Structural Biology*. Accepted.

RESEARCH INTERESTS

Protein Engineering, Design Models & Functional Evolution

My research interests center on **computational protein engineering**, especially model-driven protein design, virtual screening, evolutionary analysis, and functional redesign.

- **Protein Design Models:** I am interested in developing generative and predictive models for protein sequence, structure, and function. By integrating structural priors, evolutionary information, and experimental feedback, I hope to improve the reliability of computational protein design.
- **Virtual Screening & Functional Engineering:** I aim to build efficient screening pipelines for protein and peptide candidates, and to use evolutionary and functional analyses to guide rational modification of enzymes, binders, and other engineered proteins.

HONORS AND AWARDS

WeiGuang Program, Individual Excellence Award Jun 2025
An undergraduate research program initiated by the college to foster research skills.

Gold Award for the Practical Detachment, Tanwei College Jan 2025
Awarded for outstanding performance as the Team Leader of the 'Zhi-Shen-Qing-Lu' Social Practice delegation.

Comprehensive Excellence Award Nov 2024
Tsinghua University, University Scholarship (TOP 20%).

Gold Medal, iGEM Competition Oct 2024
The project was awarded in the premier international synthetic biology competition.

SELECTED PROJECTS

Protein-Peptide Interaction Prediction via Generative Docking & GNNs Course Project

Developed a generative-discriminative framework coupling diffusion-based docking (RAPiDock) with a Transformer-GNN scorer (ITN). Implemented multi-instance learning on 3D bipartite graphs to enable structure-aware, interpretable binding prediction. Outperformed sequence-based baselines in AUC and enrichment on pMHC I and SH3-peptide systems.

SKILLS

Programming	Python, C++, R, Java, LaTeX
Frameworks	PyTorch
Languages	English (TOEFL 94)